

Mini Project Report

Role of AI in Autonomous Robots

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1. Introduction

Autonomous robots are machines that can sense their surroundings, make decisions and carry out tasks with little or no human intervention. What separates a truly autonomous robot from a simple programmed machine is the presence of artificial intelligence: instead of following a fixed sequence of instructions, the robot interprets data coming from its sensors, reasons about the situation it is in, and chooses an action accordingly. This report looks at the role AI plays inside such robots, the main techniques involved, where they are already being used, and the obstacles that still stand in the way of wider adoption.

The interest in this topic comes from how quickly the line between 'automated' and 'autonomous' machines has shifted. A robotic arm that repeats the same weld on a car body is automated, but a delivery robot that has to choose its own path around pedestrians on a street it has never driven before is autonomous, and that difference is almost entirely due to AI.

2. What Makes a Robot Autonomous

An autonomous robot generally needs to complete three things on its own, in a loop that repeats many times per second:

1. Perceive — build an understanding of the environment from camera, lidar, radar, ultrasonic or other sensor data.
2. Decide — use that understanding to plan a path, choose an action, or predict what might happen next.
3. Act — convert the decision into motor commands while continuing to monitor the result.

AI is what allows the first two steps to work in environments that were never explicitly described to the robot beforehand. Classical automation can handle a fixed factory cell where every object is in a known position; AI is what makes it possible to handle a warehouse floor, a hospital corridor or a public road, where the layout changes from one moment to the next.

3. Core AI Techniques Used in Autonomous Robots

3.1 Computer Vision

Cameras combined with deep learning models let a robot identify objects, people, lane markings or obstacles in real time. Convolutional neural networks are the most common building block here, trained to recognize categories of objects from large sets of labeled images.

3.2 SLAM (Simultaneous Localization and Mapping)

SLAM algorithms let a robot build a map of an unfamiliar space while simultaneously figuring out its own position inside that map. This is the technique behind most vacuum robots, warehouse robots and mobile service robots that need to operate in spaces nobody pre-mapped for them.

3.3 Path Planning and Motion Control

Once the robot knows where it is and what is around it, planning algorithms (such as A*, RRT, or learned policies) compute a safe route to the goal, while motion controllers translate that route into smooth, physically realistic movement that avoids sudden or unstable actions.

3.4 Reinforcement Learning

Reinforcement learning trains a robot through trial and error inside a simulator, rewarding actions that move it closer to a goal and penalizing actions that cause failure or collision. This approach has become popular for tasks like robotic grasping and legged locomotion, where it is difficult to hand-write rules that cover every situation.

3.5 Sensor Fusion

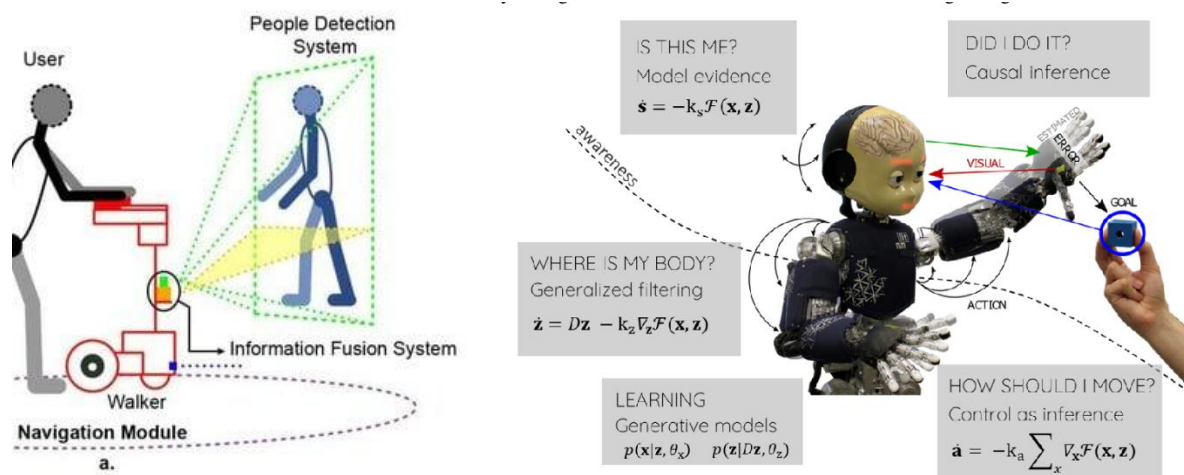
4. Applications of AI-Driven Autonomous Robots

Field	Example use of AI
Manufacturing	Vision-guided robotic arms that pick parts in random orientation, and mobile robots that navigate factory floors around moving workers.
Logistics & warehousing	Autonomous mobile robots that map a warehouse, plan the shortest route to a shelf, and avoid other robots and staff dynamically.
Self-driving vehicles	Perception and prediction models that detect lanes, pedestrians and other vehicles, and plan safe maneuvers in real traffic.
Agriculture	Robots that use vision to identify ripe fruit, distinguish crops from weeds, and adjust spraying or harvesting accordingly.
Healthcare	Surgical-assist robots that stabilize instrument movement, and hospital delivery robots that navigate corridors and elevators on their own.
Search & rescue	Drones and ground robots that use vision and mapping to explore unstable or hazardous environments without a human operator close by.



5. Example: AI in Warehouse Robots

A practical example that ties the techniques above together is a modern warehouse mobile robot. When it is first deployed, it drives through the warehouse while building a map using SLAM. During normal operation, its onboard computer continuously merges camera and lidar data through sensor fusion to detect shelves, pallets, staff and other robots. A path-planning algorithm calculates the most efficient route to the next pick location while avoiding obstacles, and if a person steps into its path, the robot's perception system recognizes this within a fraction of a second and triggers a safe stop or reroute. None of this would be possible with a fixed, pre-programmed path, since the layout of a busy warehouse changes constantly throughout the day.



6. Challenges and Limitations

- Reliability in edge cases — AI models trained on large datasets can still fail on rare situations that were not well represented during training, such as unusual obstacles or extreme weather.
- Computing power and energy — running vision and planning models in real time requires onboard processing power, which competes with battery life on mobile robots.
- Safety certification — proving that a learned, non-deterministic system behaves safely in every situation is far harder than certifying a fixed, rule-based controller.
- Cost — high-quality sensors (lidar in particular) and the compute hardware needed to run AI models in real time add significant cost to a robot.
- Data dependence — the accuracy of perception and decision-making is only as good as the data the models were trained on, so biased or limited training data leads to biased or limited robot behavior.

7. Future Trends

- Edge AI — smaller, more efficient models running directly on the robot instead of depending on a cloud connection, reducing latency and improving reliability.

- Multi-robot coordination — fleets of robots that share map and task information with each other, rather than operating in isolation.
- Human-robot collaboration — robots that can interpret natural language instructions and work safely alongside people in shared spaces, rather than being fenced off.
- Simulation-to-real transfer — training robots almost entirely in simulation and transferring the learned behavior to physical hardware with minimal retuning, cutting development time and risk.

8. Conclusion

Artificial intelligence is the component that turns a robot from a machine that repeats fixed motions into one that can operate in environments that are unpredictable and constantly changing. Computer vision, SLAM, path planning, reinforcement learning and sensor fusion each solve a different part of the perceive-decide-act loop, and together they are what make today's warehouse robots, self-driving vehicles and service robots possible. The technology still faces real limits around reliability, cost and safety certification, but the direction is clear: as AI models become more efficient and better tested, autonomous robots are likely to move from controlled environments such as factories and warehouses into more open, human-shared spaces.

9. References

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3. "Artificial Intelligence in Robotics: From Automation to Autonomous Systems," ResearchGate, 2023.